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Profile

Senior ML engineer with 10 years across academia and industry. Career arc spans wet-lab biology, computational neuroscience, and production ML. Track record owning end-to-end ML work: data engines for long-tailed open-world inference, distributed training infrastructure on Ray, and ML reliability under real-world distribution shift. Currently working on scalable distributed-AI workloads and data curation in the LLM and RL post-training space at Anyscale.

Machine Learning Experience

Machine Learning Engineer

[Anyscale](#)

San Francisco, CA

Jan. 2025 — Now

Anyscale builds Ray, the open-source distributed compute framework underlying RL post-training, large-scale training, and inference at frontier labs.

- Conceived, pitched, and leading **Ray-bench**: a benchmark for evaluating LLM agents on distributed-systems code (framework translation, Ray code generation from spec, bug fixing, code optimization), paired with **RayGym**, a sandboxed multi-node execution harness for code-RL post-training. Strategic goal: drive Ray adoption among coding agents by giving model providers a verifiable-rewards benchmark to train on. Currently leading data generation: scraping, LLM-assisted synthesis with rejection sampling, AST-based Ray-API verification, and decontamination against pre-training corpora.
- Conceived, pitched, and designing a separate RL environment to train a Ray Core resource auto-calibration model.
- Engineered scalable distributed Ray implementations of: GRPO-based RL post-training for LLMs, LLM-as-judge data curation, 670B-parameter LLM serving, multimodal training, and agentic workflows.
- Designed benchmarks comparing performance and cost across Ray OSS and RayTurbo.
- Built agentic systems for automated debugging and codebase-wide standards enforcement.
- Early user and feedback partner for Ray LLM and Ray Direct Transport (RDMA-based tensor transfer for disaggregated inference/training); surfaced bugs and informed design.

Lead Machine Learning Engineer

[Dendra Systems](#)

Remote

Feb. 2020 — Jan. 2025

Founding ML lead. Built end-to-end ML systems for large-scale, long-tailed visual recognition across thousands of classes — data engine, training infrastructure, evaluation, and serving. Domain: aerial imagery for ecosystem monitoring.

- **Data engine for long-tailed, open-world inference across thousands of target classes.**

- Active learning to systematically harvest high-leverage training data, preventing model hallucinations on out-of-distribution inputs.
- 60% training-data reduction at no performance cost via a novelty-maximizing data-pruning method I developed (original research, unpublished); yielded pareto-optimal (exponential) scaling laws.
- Few-shot models paired with an active-learning data-harvesting UI to onboard new classes and domains in hours instead of months (Gradio).
- Unsupervised “trip-wires” for detecting model hallucinations in production; alerts routed to annotation prioritization workflow (Jira).
- Annotation QA/QC: systematically surfaced mislabeled and partially labeled samples to create a “self-healing” training dataset.
- Translated SOTA self-supervised learning research into production to improve robustness and reduce labeled-data requirements (PyTorch).
- System design for strategic labeling workflows (PlantUML, *C4 diagrams*); led vendor assessment and selection for data-curation tooling.
- Authored data-collection campaigns to patch training-data biases identified through model failure analysis.

- **Training and inference infrastructure (Ray-native, cloud-scale).**

- Distributed, scale-agnostic infrastructure for training, hyperparameter tuning, and inference (Ray/Anyscale, AWS Batch).
- 20× training-cost reduction via Bayesian hyperparameter tournaments with aggressive early stopping (Ray Tune, HyperOpt, ASHA).
- Profiled and optimized multi-GPU throughput (Grafana).
- End-to-end automation of training/inference pipelines (custom orchestrator); later re-implemented serverless for reliability and cost (Step Functions, API Gateway, λ , EventBridge).
- Researched, experimented, and productionized novel ML techniques: models, samplers, optimization functions, augmentations.

- **MLOps and reliability.**

- Sanity checks for detecting silent failures during training; stratified validation across data slices.
- Debugged model failures using interpretability methods (GradCAM, transformer attention maps) and visualization of training dynamics.
- Full artifact lineage and reproducibility (MetaFlow).
- Observability across pipelines (Cloudwatch, Slack bots, UMAP, Sentry); weekly metric reviews to prevent customer-impacting incidents.
- Code quality enforcement: pre-commit hooks, CI (Bitbucket Pipelines), property-based testing (Hypothesis), runtime validation (Pandera), design-by-contract (beartype).
- 80× acceleration in insights delivery via model-in-the-loop annotation tooling.

- **ML leadership.** Founding ML lead; drove transformation from services to ML-product company. Owned ML roadmap, KPIs, and OKRs aligned with product and operations. ML automation increased throughput and profitability and unlocked new markets. Over-saw data-collection team and ran internal “ML University” lectures.

Lead Data Scientist

[PaceMate](#)

Remote

Jan. 2019 — Dec. 2019

PaceMate monitors transmissions from bluetooth-enabled heart implants, identifying life-threatening arrhythmias and alerting emergency services. Solo ML hire; took cardiac arrhythmia detection from zero to deployed deep-learning system in a regulated, life-critical pipeline — end-to-end ownership across data, modeling, clinician tooling, and security.

- Trained SOTA deep neural network for arrhythmia classification, tuned to the implant device used by the majority of patients (Keras).
- Built ECG data-processing pipelines for highly imbalanced clinical data (imbalanced-learn, custom tools).
- Built clinician-facing labeling and prediction-review dashboard for electrophysiologists (Plotly Dash).
- Built EMR analytics dashboard with performant ETL across heterogeneous medical records to enable data-driven decision-making (SQL, PySpark, spaCy, Plotly Dash).
- Upheld SOC2 standards: encryption at rest, traffic tunnelling, instance hardening.
- Translated clinical requirements into ML specs with cardiologists; presented to C-suite and met with investors.

Data Scientist

New College of FL, [F.A.R. Institute](#)

Sarasota, FL

Aug. 2018 — Dec. 2018

The Florence A. Rothman Institute supports innovation in medical data analysis. Semester-long master’s capstone project supervised by Dr. Pat McDonald. Unpaid.

- Data-driven prediction of 30-day readmission using visit clustering.
 - visit2vec: reduce high-dimensional patient visit data into low-dimensional embeddings using a technique inspired by word2vec (TensorFlow).
 - Explored structure in patient visits data by clustering patient visits using t-SNE.
- Modeled patient trajectories on years of heart failure patients from Sarasota Memorial Hospital.
 - Clustered patients over time based on cardiac and non-cardiac chronic conditions (SQL, Pandas, PySpark).
 - Created network graphs characterizing interactions between multiple chronic conditions and heart failure and their effect on mortality (NetworkX)
 - Used finite state modeling to quantify interaction between chronic conditions and mortality (PySpark, Numpy).

Research Intern

Peng Lab, [Allen Institute for Brain Science](#)

Seattle, WA

June 2018 — Aug. 2018

The neuromorphology lab investigates the architecture of the brain at the population and single-cell level. Proposed a method that would automate the biggest bottleneck to high-throughput neural cell morphological analysis.

- Deep reinforcement learning for tracing neural structures in petabytes of noisy fluorescent microscope data ([Profundo writeup](#)).
 - Implemented proof-of-concept Deep Q Network using 3D convolutions to trace neural cell structures (TensorFlow, rl-medical).
 - Created simulation environment and reward system for training agents (Matplotlib, OpenAI Gym) based on [manually traced microscope images](#).

Research Assistant

Fairhall Lab, [University of Washington](#)

Seattle, WA

Oct. 2014 — Jan. 2016

Computational neuroscience lab investigating the biophysics of neural cells. I developed agent-based dynamical models of mosquito thermal plume navigation behavior.

- Computed and visualized flight kinematic statistics and thermal sensing statistics using windtunnel flight data (Numpy, Seaborn, `scipy[interpolate, spatial, stats]`, `sklearn`, `statsmodels`).
- Formulated biophysical models of mosquito thermonavigation: applied numerical optimization algorithms to fit model to experimental data (`scipy[optimize]`, `Pandas`).
- Created animations of thermal plume navigation models (Matplotlib 3D, MayaVi).

Molecular Biologist

Various

Various

Pre-2014

Before transitioning to AI/ML, I was a molecular biology researcher. See GScholar for details.

Education

M.S. Data Science

New College of Florida

Sarasota, FL

Aug. 2017 — Dec. 2018

B.A., Chemistry/Biology (with honors)

New College of Florida

Sarasota, FL

Aug. 2007 — May 2011

Early admission (admitted 16 yrs old)

Harriet L. Wilkes Honors College

Jupiter, FL

Jul. 2006 — May 2007

Publications, Presentations, & Teaching

* 2025 [Deploy DeepSeek-R1 with vLLM and Ray Serve on Kubernetes](#) — end-to-end deployment guide for distributed serving of a 670B-parameter reasoning model.

* 2025 [Ray Direct Transport: RDMA Support in Ray Core](#) — technical write-up on RDMA-based tensor transfer for disaggregated inference/training.

- * ongoing [Personal blog](#) (80+ technical notes on Ray, LLMs, RL, ML internals, OSS projects).
Selected: [Self-authoring LLM knowledge base](#), [Using DSPy with Ray Data](#), [Slouchless: VLM for posture correction](#).
- * 2025 Speaker at Ray meetups: [Daft+Ray](#), [vLLM+Ray](#), [Skypilot+Ray](#).
- * 2024 Talk at *AI In Production Conference* on [data curation](#).
- * 2021 Invited talk at *Ray Summit*: [How Ray and Anyscale Make it Easy to do Massive-scale ML on Aerial Imagery](#) ([accompanying blog](#)).
- * 2019 Seminar at *New College of FL: Remote Sensing of Cardiac Arrhythmia at Scale using Deep Learning*.
- * 2019 Seminar at *Escuela Secundaria Tecnica de Torquinst: Inteligencia Artificial*.
- * 2018 Classroom mentor for *Udacity's Intro to Programming Nanodegree: Python for Data Analysis Track* (1-on-1 tutoring, code reviews).
- * 2015 UW Outreach: various educational events for students from low socioeconomic backgrounds.
- * 2012 [Two peer-reviewed journal articles](#) in *Genetics* and *PNAS*, also presented as posters at three national conferences.
- * 2007 Undergraduate honors thesis on RNA interference mechanisms in *C. elegans*.

Selected Awards & Grants

National Institutes of Health PA-12-149 Federal grant	2014 — 2016
<i>Self-funded grant covering salary and expenses at the UW Dept. of Biophysics.</i>	
NCF Data Scholar	2017 — 2018
<i>Full tuition waiver for master's program.</i>	
Florida "Bright Futures" Scholar	2007 — 2011
<i>Merit-based scholarship. Full undergraduate tuition.</i>	